

#. A Framework is a structured way of organizing your automation code, test data, reports and reusable components so that tests are easier to maintain, reuse and scale.

#. Problems:-

- Code Duplication**
- Hard to maintain**
- Test Data mixed with code.**
- Difficult to scale for large projects.**

Project

--tests

--test_login.py

--test_search.py

-- pages

--login_page.py

-- dashboard_search_page.py

--- data

--testdata.xlsx

--utilities

--logger.py

--config.txt

--reports

Benefits

- #. Reusable Code**

- #. Easier maintenance**

- #. Better Reporting**

- #. Data Separated from code**

- #. Team Collaboration becomes easier.**

1. POM (Page Object Model)

- #. Each page of the application gets its own class.**

Benefits:-

- #. Locaters stored in one place.**

- #. Easy Maintenance**

- #. Industry Standard**

2. Data-Driven Framework

- #. Test Data is stored externally. (Excel, CSV, JSON etc...)**

- #. Same script runs multiple data sets.**

Keyword-Driven Framework:-

#. Actions are stored as keywords in Excel.

#. Keyword Driven Testing is a framework where test actions are represented by keywords stored in external file(Excel,CSV, JSON etc...)

#. The Automation Script reads the keywords and execute the corresponding selenium action.

Benefits:-

#. Less Coding required for test cases creation

#. Test Cases can be maintained by non programmers

#. Easy to update.

#. What are all primary datatypes in Python.

#. What are all Advance datatypes in python.

#. What is the difference between a list and tuple.

#. What is the difference between list and set.

#. What are mutable and immutable objects.

#. What is slicing

#. What is dictionary.

#. What is decorator

#. What is a closure.

#. What are all the four pillars of OOPS.

#. What is Inheritance

#. What is Polymorphism

#. What is Encapsulation

#. What is Abstraction

#. What is method overloading and overriding.

#. What is Method Resolution Order (MRO)

#. What are all Python magic(dunder) methods.